Shrimayee Umesh Deshpande

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Summary

Software engineer with 2+ years of experience building scalable backend systems and automation platforms at BlackRock. Expertise in Python, Go, and SQL to design distributed data pipelines and real-time monitoring tools, reducing manual audit effort by over 12,000 hours annually. Skilled in architecting high-performance services with CI/CD, cloud deployment on Azure and AWS, and collaborating across teams to drive innovation.

Education

University of Florida — **Aug 2025 - May 2027**
Masters of Science, Artificial Intelligence Systems — *FL*
- **GPA:** 3.77

Vellore Institute of Technology, Chennai — **Jun 2019 - May 2023**
Bachelor of Technology, Electronics & Computer Engineering — *Chennai*
- **GPA:** 8.66

Skills

- **Programming & Frameworks**: Python, C/C++, MATLAB, SQL, Go, Perl, HTML/CSS, JavaScript (React), Flask, Django, FastAPI, Streamlit, Gradio, REST APIs
- **Data & Machine Learning**: Data Analysis, Data Science, Machine Learning, SLMs, Transformers, Deep Learning, Trustworthy Machine Learning, Model Evaluation, scikit-learn, PyTorch, Tensorflow OpenCV, Artificial Intelligence
- **Cloud, DevOps & Monitoring**: Azure, AWS, Docker, Prometheus, Grafana, Kubernetes, Git, CI/CD, Azure DevOps, Ansible, AWX, Shell Scripting
- **Systems & Design**: Distributed Systems, Data Structures, Algorithms, Software Design
- **Professional Skills**: Leadership, Problem Solving, Analytical Thinking, Cross-team Collaboration, Ownership

Work Experience

BlackRock — **Jul 2023 - Aug 2025**
Analyst Software Engineer
Discrepancy Audit & Configuration Drift Monitoring
- Designed and owned DART (Discrepancy Analysis & Remediation Tool), an end-to-end Python-based analytics platform built using Pandas and optimized SQL pipelines to monitor 7,000+ application servers across 400 client environments; implemented a discrepancy analyzer powered by machine learning and artificial intelligence, real-time dashboard, and auto-remediation workflows, eliminating 6 FTE (~12,000+ hours/year) of manual audit effort.
- Engineered data processing pipelines and complex SQL queries using optimized data structures and algorithms to detect configuration drift in production, enabling real-time alerting and visualization through a frontend dashboard and Grafana; significantly improved incident detection speed and reduced manual debugging effort.
- Built Python and Java-based onboarding services integrated with Azure DevOps CI/CD pipelines, implementing 60+ static and dynamic validation checks; reduced application server onboarding time by 60% and improved compliance and standardization across environments.
- Developed and optimized Ansible AWX workflows for large-scale client builds across 200+ nodes, enabling parallel execution and self-service onboarding; increased automation from 60% to 95% and reduced operational turnaround time.
- Designed scalable distributed system architecture using software design best practices for future Kubernetes integration and collaborated cross-functionally to move solutions from design to production, enabling smoother deployments and supporting growth
- Led a global team to design and deliver interactive Power BI dashboards for financial portfolio analytics, modeling DoD, MtD, and YtD active duration and curve metrics from historical data using Python pipelines.

Projects

Sprint Insights Generator
AgileAI
- Experimented with and optimized GPT-2/3.5, LLaMA-3, and BART on 30 multi-domain projects for structured text generation of Agile Epics/Features/User Stories, improving quality by +0.06 BLEU and +0.05 ROUGE-L, and reducing hallucinations by 30% via template grounding and hierarchy validation.
- Delivered a Dockerized microservices architecture with concurrent processing of Azure DevOps API calls, monitored via Prometheus and Grafana, and built a Gradio UI that applies a regression-based algorithm for story point estimation and automated ticket assignment from historical sprint data.

SLM Ethical Benchmarking & Storytelling Edge Device
- Benchmarking TinyLLaMA, Phi-3, Gemma, and Mistral 7B across ethical and reliability dimensions including bias categories, AI governance compliance, hallucination detection, sentiment analysis, and data privacy leak rates, to develop generalized guardrails for safe deployment of Small Language Models.
- Built a scalable benchmarking pipeline on Kubernetes and Docker to evaluate SLM robustness, and implemented a C++ TinyStories assistant on ESP32 using FreeRTOS for multithreaded on-device inference, ensuring safe, interactive story generation.

Learning Functional Connectivity Representations for Autism from fMRI
NeuroAlign
- Built an end-to-end neuroimaging ML pipeline on the ABIDE dataset, extracting ROI-based functional connectivity features (~6K per subject) from resting-state fMRI for ASD vs control classification.
- Trained and compared Logistic Regression, SVM, and a PyTorch MLP for representation learning, achieving ~0.78 ROC-AUC and demonstrating primarily linear separability in high-dimensional brain connectivity data.

ER Optimization Using Healthcare and Social Determinants Data
DataFest
- Analyzed large-scale healthcare and social determinants data from DataFest using JMP for preprocessing and PowerBI to build interactive dashboards assessing ER utilization, capacity, and geographic demand patterns.
- Applied data-driven analysis to identify key factors driving avoidable ER visits and uncovered survey bias (~20% response rate), highlighting underrepresented populations and informing equitable healthcare resource allocation